

- Description

This module manages a smart temperature-controlled logistics system, ensuring that:

1. **Shipments are continuously monitored for temperature compliance** based on predefined thresholds.
2. **If the temperature of a shipment deviates from the acceptable range, an alert is triggered.**
3. **Automatic cooling or heating adjustments are applied to maintain the required conditions.**
4. **A log of all temperature fluctuations and corrective actions** is maintained for quality control.

This system ensures that perishable goods, such as food or pharmaceuticals, remain within safe temperature limits throughout transit.

- Specification

module SmartTemperatureControl

var currentTemperature, minTemp, maxTemp: real

var temperatureLog: sequence of string

process updateTemperature(newTemp: real) flag: boolean

functional scenario form:

newTemp < minTemp && flag = true and temperatureLog = temperatureLog

⊕ ("Heating system activated."), Temperature too low! Heating activated.

||

newTemp > maxTemp && flag = true and temperatureLog = temperatureLog

⊕ ("Cooling system activated."), Temperature too high! Cooling activated.

||

newTemp <= maxTemp && Temperature stable, no action required.

end process

end module

- Java Code

```
package code;
import java.util.*;
class Shipment {
    private String shipmentId;
    private double currentTemperature;
    private final double minTemp;
    private final double maxTemp;
    private List<String> temperatureLog;
    public Shipment(String shipmentId, double initialTemp, double minTemp, double maxTemp) {
        this.shipmentId = shipmentId;
        this.currentTemperature = initialTemp;
        this.minTemp = minTemp;
```

```

        this.maxTemp = maxTemp;
        this.temperatureLog = new ArrayList<>();
        logTemperature("Initialized with temperature: " + initialTemp);
    }
    public void updateTemperature(double newTemp) {
        currentTemperature = newTemp;
        logTemperature("Updated temperature: " + newTemp);
        if (newTemp < minTemp) {
            triggerHeating();
        } else if (newTemp > maxTemp) {
            triggerCooling();
        } else {
            System.out.printf("Shipment %s is within the safe temperature range: %.1f°C\n", shipmentId, newTemp);
        }
    }
    private void triggerHeating() {
        System.out.printf("ALERT! Shipment %s too cold (%.1f°C). Activating heating system!\n", shipmentId, currentTemperature);
        logTemperature("Heating system activated due to low temperature.");
    }
    private void triggerCooling() {
        System.out.printf("ALERT! Shipment %s too hot (%.1f°C). Activating cooling system!\n", shipmentId, currentTemperature);
        logTemperature("Cooling system activated due to high temperature.");
    }
    private void logTemperature(String message) {
        temperatureLog.add(message);
    }
    public void printTemperatureLog() {
        System.out.println("\nTemperature Log for Shipment " + shipmentId + ":");
        for (String log : temperatureLog) {
            System.out.println(log);
        }
    }
}
public class c17_SmartLogisticsSystem {
    public static void main(String[] args) {
        Shipment vaccineShipment = new Shipment("VX123", 5.0, 2.0, 8.0);
        // testcase-VT:

```

```
// vaccineShipment.updateTemperature(-6.96);
// vaccineShipment.updateTemperature(23.37);
// vaccineShipment.updateTemperature(32.56);
// vaccineShipment.updateTemperature(31.16);
// vaccineShipment.updateTemperature(27.84);
// vaccineShipment.updateTemperature(-9.85);
// vaccineShipment.updateTemperature(11.33);
// vaccineShipment.updateTemperature(-12.49);
// vaccineShipment.updateTemperature(-16.02);
// vaccineShipment.updateTemperature(15.64);
// vaccineShipment.updateTemperature(38.35);
// vaccineShipment.updateTemperature(-13.78);
// vaccineShipment.updateTemperature(26.43);
// vaccineShipment.updateTemperature(-3.70);
// vaccineShipment.updateTemperature(39.56);
// vaccineShipment.updateTemperature(-16.36);
// vaccineShipment.updateTemperature(-19.30);
// vaccineShipment.updateTemperature(25.04);
// vaccineShipment.updateTemperature(10.37);
// vaccineShipment.updateTemperature(31.94);
// vaccineShipment.updateTemperature(-17.07);
// vaccineShipment.updateTemperature(-6.95);
// vaccineShipment.updateTemperature(5.43);
// vaccineShipment.updateTemperature(-15.54);
// vaccineShipment.updateTemperature(13.09);
// vaccineShipment.updateTemperature(17.22);
// vaccineShipment.updateTemperature(30.95);
// vaccineShipment.updateTemperature(39.57);
// vaccineShipment.updateTemperature(3.12);
// vaccineShipment.updateTemperature(3.49);
// vaccineShipment.updateTemperature(-5.33);
// vaccineShipment.updateTemperature(18.01);
// vaccineShipment.updateTemperature(-10.39);
// vaccineShipment.updateTemperature(13.63);
// vaccineShipment.updateTemperature(29.32);
// vaccineShipment.updateTemperature(-11.75);
// vaccineShipment.updateTemperature(4.86);
// vaccineShipment.updateTemperature(6.04);
// vaccineShipment.updateTemperature(18.57);
// vaccineShipment.updateTemperature(7.12);
// vaccineShipment.updateTemperature(17.73);
// vaccineShipment.updateTemperature(13.07);
// vaccineShipment.updateTemperature(-12.78);
// vaccineShipment.updateTemperature(-9.24);
```

```
// vaccineShipment.updateTemperature(-2.05);
// vaccineShipment.updateTemperature(33.58);
// vaccineShipment.updateTemperature(-1.01);
// vaccineShipment.updateTemperature(-0.65);
// vaccineShipment.updateTemperature(8.85);
// vaccineShipment.updateTemperature(-7.23);
// vaccineShipment.updateTemperature(6.32);
// vaccineShipment.updateTemperature(-2.13);
// vaccineShipment.updateTemperature(-12.56);
// vaccineShipment.updateTemperature(10.50);
// vaccineShipment.updateTemperature(29.41);
// vaccineShipment.updateTemperature(0.44);
// vaccineShipment.updateTemperature(6.50);
// vaccineShipment.updateTemperature(0.01);
// vaccineShipment.updateTemperature(4.64);
// vaccineShipment.updateTemperature(2.35);
// vaccineShipment.updateTemperature(-16.47);
// vaccineShipment.updateTemperature(8.77);
// vaccineShipment.updateTemperature(14.64);
// vaccineShipment.updateTemperature(19.57);
// vaccineShipment.updateTemperature(-4.46);
// vaccineShipment.updateTemperature(30.93);
// vaccineShipment.updateTemperature(23.37);
// vaccineShipment.updateTemperature(36.24);
// vaccineShipment.updateTemperature(-13.59);
// vaccineShipment.updateTemperature(26.36);
// vaccineShipment.updateTemperature(-5.58);
// vaccineShipment.updateTemperature(14.73);
// vaccineShipment.updateTemperature(15.87);
// vaccineShipment.updateTemperature(23.74);
// vaccineShipment.updateTemperature(36.38);
// vaccineShipment.updateTemperature(11.36);
// vaccineShipment.updateTemperature(19.65);
// vaccineShipment.updateTemperature(-0.38);
// vaccineShipment.updateTemperature(-2.81);
// vaccineShipment.updateTemperature(-9.99);
// vaccineShipment.updateTemperature(32.82);
// vaccineShipment.updateTemperature(38.00);
// vaccineShipment.updateTemperature(18.74);
// vaccineShipment.updateTemperature(4.80);
// vaccineShipment.updateTemperature(-3.95);
// vaccineShipment.updateTemperature(33.40);
// vaccineShipment.updateTemperature(32.82);
// vaccineShipment.updateTemperature(0.55);
```

```
// vaccineShipment.updateTemperature(27.48);
// vaccineShipment.updateTemperature(-14.25);
// vaccineShipment.updateTemperature(18.40);
// vaccineShipment.updateTemperature(19.88);
// vaccineShipment.updateTemperature(-12.70);
// vaccineShipment.updateTemperature(-13.24);
// vaccineShipment.updateTemperature(-15.82);
// vaccineShipment.updateTemperature(8.20);
// vaccineShipment.updateTemperature(14.46);
// vaccineShipment.updateTemperature(-2.63);
// vaccineShipment.updateTemperature(16.14);
// vaccineShipment.updateTemperature(-9.22);

//testcase-FT:
// vaccineShipment.updateTemperature(1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(9.452954591057998E290);
// vaccineShipment.updateTemperature(2.4233672352148374E293);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(9.063142030601998E290);
// vaccineShipment.updateTemperature(2.3292275018647134E293);
// vaccineShipment.updateTemperature(-1.7976931348617194E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(1.3740892756073996E291);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(1.3740892756073996E291);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(3.76498710603488E303);
// vaccineShipment.updateTemperature(4.765397965638438E304);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(-6.556292609497859E307);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(9.063142030601998E290);
// vaccineShipment.updateTemperature(1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(9.063142030601998E290);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(7.758198681499685E295);
```

```
// vaccineShipment.updateTemperature(7.758198681499685E295);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(9.258048310829998E290);
// vaccineShipment.updateTemperature(3.976301471980288E300);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(9.355501450943998E290);
// vaccineShipment.updateTemperature(6.153518709348742E295);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(-4.153611213935024E307);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(4.580297585357999E290);
// vaccineShipment.updateTemperature(2.3746406651578374E293);
// vaccineShipment.updateTemperature(3.919626690863351E295);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(1.153611213935024E307);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(-0.0);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(9.063142030601998E290);
// vaccineShipment.updateTemperature(-1.7976931348623141E308);
// vaccineShipment.updateTemperature(9.355501450943998E290);
// vaccineShipment.updateTemperature(-4.9E-324);
// vaccineShipment.updateTemperature(1.7976931348623157E308);
// vaccineShipment.updateTemperature(-4.9E-324);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(1.7976931348623157E308);
// vaccineShipment.updateTemperature(1.7976931348623157E308);
// vaccineShipment.updateTemperature(-4.9E-324);
// vaccineShipment.updateTemperature(7.758198681499685E295);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(9.258048310829998E290);
```

```
// vaccineShipment.updateTemperature(3.976301471980288E300);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(9.355501450943998E290);
// vaccineShipment.updateTemperature(6.153518709348742E295);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(-4.153611213935024E307);
// vaccineShipment.updateTemperature(0.0);
// vaccineShipment.updateTemperature(4.580297585357999E290);
// vaccineShipment.updateTemperature(2.3746406651578374E293);
// vaccineShipment.updateTemperature(1.3740892756073996E291);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(1.3740892756073996E291);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(3.76498710603488E303);
// vaccineShipment.updateTemperature(4.765397965638438E304);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(1.3740892756073996E291);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(1.3740892756073996E291);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(3.76498710603488E303);
// vaccineShipment.updateTemperature(4.765397965638438E304);
// vaccineShipment.updateTemperature(-1.7976931348623157E308);
// vaccineShipment.updateTemperature(-4.9E-324);
```

```
// testcase-Z3:
```

```
// vaccineShipment.updateTemperature(2.37);
// vaccineShipment.updateTemperature(1.37);
// vaccineShipment.updateTemperature(-1.78);
// vaccineShipment.updateTemperature(1.37);
// vaccineShipment.updateTemperature(-1.78);
// vaccineShipment.updateTemperature(3.763);
// vaccineShipment.updateTemperature(4.7);
// vaccineShipment.updateTemperature(-1.08);
// vaccineShipment.updateTemperature(1.31);
// vaccineShipment.updateTemperature(-1.08);
// vaccineShipment.updateTemperature(1.31);
// vaccineShipment.updateTemperature(2.37);
// vaccineShipment.updateTemperature(1.37);
// vaccineShipment.updateTemperature(-1.78);
// vaccineShipment.updateTemperature(1.37);
// vaccineShipment.updateTemperature(-1.78);
```

[illegible]

```
// vaccineShipment.updateTemperature(3.763);
// vaccineShipment.updateTemperature(4.7);
// vaccineShipment.updateTemperature(-1.08);
// vaccineShipment.updateTemperature(1.31);
// vaccineShipment.updateTemperature(-1.08);
// vaccineShipment.updateTemperature(1.31);
// vaccineShipment.updateTemperature(2.37);
// vaccineShipment.updateTemperature(1.37);
// vaccineShipment.updateTemperature(-1.78);
// vaccineShipment.updateTemperature(1.37);
// vaccineShipment.updateTemperature(-1.78);
// vaccineShipment.updateTemperature(3.763);
// vaccineShipment.updateTemperature(4.7);
// vaccineShipment.updateTemperature(-1.08);
// vaccineShipment.updateTemperature(1.31);
// vaccineShipment.updateTemperature(-1.08);
// vaccineShipment.updateTemperature(1.31);
// vaccineShipment.updateTemperature(2.37);
// vaccineShipment.updateTemperature(1.37);
// vaccineShipment.updateTemperature(-1.78);
// vaccineShipment.updateTemperature(1.37);
// vaccineShipment.updateTemperature(-1.78);
// vaccineShipment.updateTemperature(3.763);
// vaccineShipment.updateTemperature(4.7);
// vaccineShipment.updateTemperature(-1.08);
// vaccineShipment.updateTemperature(2.37);
// vaccineShipment.updateTemperature(1.37);
// vaccineShipment.updateTemperature(-1.78);
// vaccineShipment.updateTemperature(1.37);
// vaccineShipment.updateTemperature(-1.78);
// vaccineShipment.updateTemperature(3.763);
// vaccineShipment.updateTemperature(4.7);
// vaccineShipment.updateTemperature(-1.08);
// vaccineShipment.updateTemperature(1.31);
// vaccineShipment.updateTemperature(-1.08);
// vaccineShipment.updateTemperature(1.31);
// vaccineShipment.updateTemperature(-1.08);
// vaccineShipment.updateTemperature(3.7);
// vaccineShipment.updateTemperature(4.74);
// vaccineShipment.updateTemperature(-1.308);
// vaccineShipment.updateTemperature(-1.78);

// testcase-UVT:
vaccineShipment.updateTemperature(-6.96);
vaccineShipment.updateTemperature(23.37);
```

```
vaccineShipment.updateTemperature(32.56);  
vaccineShipment.updateTemperature(31.16);  
vaccineShipment.updateTemperature(27.84);  
vaccineShipment.updateTemperature(-9.85);  
vaccineShipment.updateTemperature(11.33);  
vaccineShipment.updateTemperature(-12.49);  
vaccineShipment.updateTemperature(-16.02);  
vaccineShipment.updateTemperature(15.64);  
vaccineShipment.updateTemperature(38.35);  
vaccineShipment.updateTemperature(-13.78);  
vaccineShipment.updateTemperature(26.43);  
vaccineShipment.updateTemperature(-3.70);  
vaccineShipment.updateTemperature(39.56);  
vaccineShipment.updateTemperature(-16.36);  
vaccineShipment.updateTemperature(-19.30);  
vaccineShipment.updateTemperature(25.04);  
vaccineShipment.updateTemperature(10.37);  
vaccineShipment.updateTemperature(31.94);  
vaccineShipment.updateTemperature(-17.07);  
vaccineShipment.updateTemperature(-6.95);  
vaccineShipment.updateTemperature(5.43);  
vaccineShipment.updateTemperature(-15.54);  
vaccineShipment.updateTemperature(13.09);  
vaccineShipment.updateTemperature(17.22);  
vaccineShipment.updateTemperature(30.95);  
vaccineShipment.updateTemperature(39.57);  
vaccineShipment.updateTemperature(3.12);  
vaccineShipment.updateTemperature(3.49);  
vaccineShipment.updateTemperature(-5.33);  
vaccineShipment.updateTemperature(18.01);  
vaccineShipment.updateTemperature(-10.39);  
vaccineShipment.updateTemperature(13.63);  
vaccineShipment.updateTemperature(29.32);  
vaccineShipment.updateTemperature(-11.75);  
vaccineShipment.updateTemperature(4.86);  
vaccineShipment.updateTemperature(6.04);  
vaccineShipment.updateTemperature(18.57);  
vaccineShipment.updateTemperature(7.12);  
vaccineShipment.updateTemperature(17.73);  
vaccineShipment.updateTemperature(13.07);  
vaccineShipment.updateTemperature(-12.78);  
vaccineShipment.updateTemperature(-9.24);  
vaccineShipment.updateTemperature(-2.05);  
vaccineShipment.updateTemperature(33.58);
```

```
vaccineShipment.updateTemperature(-1.01);  
vaccineShipment.updateTemperature(-0.65);  
vaccineShipment.updateTemperature(8.85);  
vaccineShipment.updateTemperature(-7.23);  
vaccineShipment.updateTemperature(6.32);  
vaccineShipment.updateTemperature(-2.13);  
vaccineShipment.updateTemperature(-12.56);  
vaccineShipment.updateTemperature(10.50);  
vaccineShipment.updateTemperature(29.41);  
vaccineShipment.updateTemperature(0.44);  
vaccineShipment.updateTemperature(6.50);  
vaccineShipment.updateTemperature(0.01);  
vaccineShipment.updateTemperature(4.64);  
vaccineShipment.updateTemperature(2.35);  
vaccineShipment.updateTemperature(-16.47);  
vaccineShipment.updateTemperature(8.77);  
vaccineShipment.updateTemperature(14.64);  
vaccineShipment.updateTemperature(19.57);  
vaccineShipment.updateTemperature(-4.46);  
vaccineShipment.updateTemperature(30.93);  
vaccineShipment.updateTemperature(23.37);  
vaccineShipment.updateTemperature(36.24);  
vaccineShipment.updateTemperature(-13.59);  
vaccineShipment.updateTemperature(26.36);  
vaccineShipment.updateTemperature(-5.58);  
vaccineShipment.updateTemperature(14.73);  
vaccineShipment.updateTemperature(15.87);  
vaccineShipment.updateTemperature(23.74);  
vaccineShipment.updateTemperature(36.38);  
vaccineShipment.updateTemperature(11.36);  
vaccineShipment.updateTemperature(19.65);  
vaccineShipment.updateTemperature(-0.38);  
vaccineShipment.updateTemperature(-2.81);  
vaccineShipment.updateTemperature(-9.99);  
vaccineShipment.updateTemperature(32.82);  
vaccineShipment.updateTemperature(38.00);  
vaccineShipment.updateTemperature(18.74);  
vaccineShipment.updateTemperature(4.80);  
vaccineShipment.updateTemperature(-3.95);  
vaccineShipment.updateTemperature(33.40);  
vaccineShipment.updateTemperature(32.82);  
vaccineShipment.updateTemperature(0.55);  
vaccineShipment.updateTemperature(27.48);  
vaccineShipment.updateTemperature(-14.25);
```

```

        vaccineShipment.updateTemperature(18.40);
        vaccineShipment.updateTemperature(19.88);
        vaccineShipment.updateTemperature(-12.70);
        vaccineShipment.updateTemperature(-13.24);
        vaccineShipment.updateTemperature(-15.82);
        vaccineShipment.updateTemperature(8.20);
        vaccineShipment.updateTemperature(14.46);
        vaccineShipment.updateTemperature(-2.63);
        vaccineShipment.updateTemperature(16.14);
        vaccineShipment.updateTemperature(-9.22);
//        vaccineShipment.printTemperatureLog();
    }
}

```

- Coverage report

1. Test case number: 100
2. V-Method testing branch coverage: 66.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	838	842
▼ src	0.5 %	4	838	842
▼ code	0.5 %	4	838	842
▼ c17_SmartLogisticsSystem.java	66.7 %	4	2	6
> Shipment	66.7 %	4	2	6
> c17_SmartLogisticsSystem		0	0	0

3. Vibration testing instruction coverage: 94.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	2.5 %	421	16,160	16,581
▼ src	2.5 %	421	16,160	16,581
▼ code	2.5 %	421	16,160	16,581
▼ c17_SmartLogisticsSystem.java	94.6 %	421	24	445
> Shipment	84.2 %	112	21	133
> c17_SmartLogisticsSystem	99.0 %	309	3	312

4. Fuzz testing branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.4 %	3	839	842
▼ src	0.4 %	3	839	842
▼ code	0.4 %	3	839	842
▼ c17_SmartLogisticsSystem.java	50.0 %	3	3	6
> Shipment	50.0 %	3	3	6
> c17_SmartLogisticsSystem		0	0	0

5. Fuzz testing instruction coverage: 91%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	2.4 %	405	16,176	16,581
▼ src	2.4 %	405	16,176	16,581
▼ code	2.4 %	405	16,176	16,581
▼ c17_SmartLogisticsSystem.java	91.0 %	405	40	445
> Shipment	72.2 %	96	37	133
> c17_SmartLogisticsSystem	99.0 %	309	3	312

6. Z3 branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.4 %	3	837	840
✓ src	0.4 %	3	837	840
✓ code	0.4 %	3	837	840
✓ c17_SmartLogisticsSystem.java	50.0 %	3	3	6
> Shipment	50.0 %	3	3	6
> c17_SmartLogisticsSystem		0	0	0

7. Z3 instruction coverage: 89.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	1.8 %	397	21,698	22,095
✓ src	1.8 %	397	21,698	22,095
✓ code	1.8 %	397	21,698	22,095
✓ c17_SmartLogisticsSystem.java	89.2 %	397	48	445
> Shipment	66.2 %	88	45	133
> c17_SmartLogisticsSystem	99.0 %	309	3	312

8. Unimproved V-method branch coverage: 66.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.5 %	4	836	840
✓ src	0.5 %	4	836	840
✓ code	0.5 %	4	836	840
✓ c17_SmartLogisticsSystem.java	66.7 %	4	2	6
> Shipment	66.7 %	4	2	6
> c17_SmartLogisticsSystem		0	0	0

9. Unimproved V-method instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	1.9 %	421	21,674	22,095
✓ src	1.9 %	421	21,674	22,095
✓ code	1.9 %	421	21,674	22,095
✓ c17_SmartLogisticsSystem.java	94.6 %	421	24	445
> Shipment	84.2 %	112	21	133
> c17_SmartLogisticsSystem	99.0 %	309	3	312